ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 2

Micronutrients — The Unsung Heroes

Duration	55 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: Why Micronutrients Are Critical for Athletes

Vitamins and Minerals in Athletic Performance

While macronutrients provide energy and structure, micronutrients (vitamins and minerals) are the molecular tools that enable every metabolic reaction in the body. Without adequate micronutrients, macronutrient metabolism is impaired — like having excellent fuel in a car with a malfunctioning engine.

Athletes have higher micronutrient requirements than sedentary individuals because exercise increases metabolic rate, sweat losses, oxidative stress, tissue turnover, and demand for energy metabolism co-factors.

Micronutrient	RDA (general)	Sports Relevance	Best Sources
Vitamin D3	600–2000 IU (often more needed)	Testosterone, bone health, muscle function, immune	Sunlight, fatty fish, fortified dairy, D3 supplement
Magnesium	310–420mg	ATP synthesis, muscle contraction, sleep (GABA), >300 enzymes	Pumpkin seeds, spinach, dark chocolate, legumes, nut
Iron	18mg (F) / 8mg (M)	Haemoglobin, oxygen transport, endurance	Red meat, organ meat, spinach + vit C, lentils (non-hae
Zinc	8–11mg	Testosterone synthesis, immune, wound healing, taste/smell	Oysters, red meat, pumpkin seeds, legumes, hemp see
B12	2.4mcg	Nerve myelin, DNA synthesis, red blood cells, energy metabolism	Animal products, supplement essential for vegans
Calcium	1000–1200mg	Bone density, muscle contraction, nerve signalling	Dairy, fortified plant milk, ragi (finger millet), sesame
Vitamin C	75–90mg (higher for athletes)	Collagen synthesis, antioxidant, iron absorption, immune	Citrus, guava, bell peppers, amla (Indian gooseberry)
B-vitamins (B1,B2,B3,B6)	Varied	Co-factors in glycolysis and Krebs cycle — energy metabolism	Whole grains, legumes, meat, dairy, eggs

SECTION 2: Vitamin D — The Hormone Masquerading as a Vitamin

Vitamin D — Far More Than Bone Health

Vitamin D is technically a prohormone, not a vitamin — the body synthesises it from cholesterol in the skin upon UVB exposure. It is converted to its active form (calcitriol / 1,25-dihydroxyvitamin D) by the liver and kidneys.

Vitamin D receptors (VDRs) are found in virtually every cell in the body — muscle, brain, immune cells, heart, reproductive organs, bone. This explains its broad physiological effects beyond calcium metabolism.

- Bone health: regulates calcium and phosphate absorption and bone mineralisation
- Muscle function: VDR in muscle cells — low D3 associated with reduced muscle force and increased injury risk

- Testosterone: studies show inverse correlation between D3 deficiency and testosterone levels
- Immune modulation: reduces autoimmune disease risk; critical for innate immunity (COVID outcomes correlated)
- Mental health: low D3 associated with depression, seasonal affective disorder (SAD)
- Insulin sensitivity: D3 deficiency impairs insulin signalling

Deficiency in India: Paradoxically, despite abundant sunshine, D3 deficiency affects 70–90% of Indians. Reasons: indoor work, skin melanin (reduces UV synthesis), limited D3-rich foods in traditional vegetarian diets, cultural practices (covered clothing, avoiding midday sun). Testing (25-OH-D blood test) is recommended.

D3 Status	Blood Level (25-OH-D)	Clinical Significance
Deficient	<20 ng/mL (<50 nmol/L)	High risk: osteoporosis, immune dysfunction, muscle weakness
Insufficient	20–29 ng/mL	Suboptimal; many biological functions impaired
Optimal	40–60 ng/mL	Target range for athletes and health-conscious individuals
Toxic (rare)	>150 ng/mL	Hypercalcaemia; requires extremely high supplementation (>10,000 I

Supplementation: Most adults in India benefit from 2000–4000 IU Vitamin D3 daily year-round. Take with K2 (100–200mcg MK-7) to direct calcium to bones (not arteries). Fat-soluble — take with a fatty meal for optimal absorption.

SECTION 3: Iron — The Oxygen Carrier

Iron Deficiency — The Silent Performance Killer

Iron is the central atom of haemoglobin (in red blood cells) and myoglobin (in muscle cells) — the proteins that carry and store oxygen. Iron deficiency impairs oxygen delivery to working muscles, reducing VO2 max, endurance capacity, and cognitive function.

Haem vs Non-Haem Iron

Type	Sources	Absorption Rate	Enhancement/Inhibition
Haem iron	Red meat, organ meat, poultry, fish	15–35% absorbed	Not affected by phytates; best bioavailable form
Non-haem iron	Lentils, spinach, fortified cereals, tofu	2–20% absorbed	Enhanced by vitamin C (+2–4×); inhibited by calcium, tea polypher

Practical tip for vegetarians: Consume non-haem iron sources with vitamin C (lemon juice, amla, guava, bell peppers). Avoid tea/coffee within 1 hour of iron-rich meals (tannins inhibit non-haem absorption by up to 60%).

Sports anaemia: Endurance athletes (especially runners) face higher iron losses through haemolysis (foot-strike destruction of RBCs), sweat, and micro-GI bleeding. Female endurance athletes are particularly at risk — menstrual losses compound dietary shortfall.

Iron toxicity: Iron is pro-oxidant in excess — drives free radical production. Do NOT supplement iron without confirmed deficiency (serum ferritin <30 ng/mL in athletes; <20 ng/mL in sedentary). Excess iron increases cardiovascular disease risk.

SECTION 4: Magnesium — The Macro-Mineral You're Probably Deficient In

Magnesium — 300+ Enzymes, One Mineral

Magnesium is the fourth most abundant mineral in the human body and co-factor for over 300 enzymatic reactions, including all ATP-generating reactions. Without magnesium, ATP cannot be utilised — every muscle contraction, nerve impulse, and DNA synthesis requires Mg^{2}■-ATP.

- Energy metabolism: co-factor for all ATP synthesis (glycolysis, Krebs, ETC)
- Muscle function: required for calcium to be pumped out of muscle post-contraction (relaxation)
- Protein synthesis: ribosomal activity requires magnesium
- Sleep: activates GABA receptors — the brain's primary inhibitory neurotransmitter
- Blood pressure: relaxes vascular smooth muscle; deficiency is a major driver of hypertension
- Insulin sensitivity: magnesium deficiency impairs glucose transporter function (GLUT4)

Deficiency prevalence: 60–70% of people in developed countries consume below the RDA. Soil depletion, refined food consumption, high sugar/alcohol intake (all increase magnesium excretion), and gut absorption issues contribute.

Form	Bioavailability	Best For	Side Effects
Magnesium glycinate	High — gentle	Sleep, anxiety, general use	Minimal GI effects
Magnesium citrate	Good	Constipation, general deficiency	Loose stools at high dose
Magnesium malate	Good	Energy, fibromyalgia	Mild
Magnesium oxide	Low (~4%)	NOT recommended	High GI issues
Magnesium L-threonate	High (brain crossing)	Cognitive function, memory	Expensive

Food sources: Pumpkin seeds (156mg/28g), dark chocolate (50mg/28g), spinach (78mg/half cup cooked), almonds (77mg/28g), black beans (60mg/half cup), avocado (58mg per avocado), salmon (26mg/85g).

SECTION 5: Antioxidants — Managing Oxidative Stress

Free Radicals, Antioxidants, and the Exercise Paradox

Exercise generates reactive oxygen species (ROS) — chemically reactive molecules with unpaired electrons that damage cell membranes, DNA, and proteins. This is oxidative stress — and it is both a trigger for adaptation AND a potential cause of cell damage when excessive.

The hormesis paradox: Mild oxidative stress from exercise triggers antioxidant enzyme upregulation (superoxide dismutase, catalase, glutathione peroxidase) — the body's internal antioxidant army. Supplementing with high-dose exogenous antioxidants (vitamin C, E) post-workout may blunt these adaptive responses, similar to how cold water immersion blunts hypertrophy.

Antioxidant	Source	Function	Note for Athletes
Vitamin C	Citrus, guava, amla, bell peppers	Aqueous phase antioxidant; regenerates vitamin E; collagen synthesis	Safe to supplement; don't mega-dose post-workout
Vitamin E	Nuts, seeds, olive oil, avocado	Lipid phase antioxidant; protects cell membranes	Food sources preferred; synthetic form (dl-alpha) less effe
Beta-carotene (provitamin A)	Carrots, sweet potato, mango, spinach	Antioxidant; converted to retinol (vit A)	Excess beta-carotene (not retinol) safe; food over supplem
Polyphenols	Berries, green tea, turmeric, dark chocolate	Anti-inflammatory; modulate gut microbiome	Beneficial chronically; avoid very high doses acutely post-w
Glutathione	Synthesised in body; boosted by NAC supplementation	Master intracellular antioxidant	N-acetyl cysteine (NAC) is an effective precursor
Selenium	Brazil nuts, fish, eggs, meat	Glutathione peroxidase co-factor; thyroid function	1–2 Brazil nuts/day provides RDA; excess is toxic

SECTION 6: Gut Health and Microbiome Nutrition

The Gut-Muscle Axis

The gut microbiome — the trillions of bacteria, fungi, and other microorganisms in the digestive tract — influences nutrient absorption, inflammation, immune function, mental health, and increasingly, athletic performance. A diverse, healthy microbiome is associated with:

- Better short-chain fatty acid (SCFA) production → anti-inflammatory, gut barrier integrity
- Enhanced nutrient absorption (B-vitamins, vitamin K2 synthesised by bacteria)
- Reduced systemic inflammation → faster recovery from exercise
- Lower depression/anxiety risk via gut-brain axis (vagus nerve signalling)
- Better insulin sensitivity → improved glucose partitioning to muscle

Microbiome Support Strategy	Examples	Mechanism
Prebiotics (feed bacteria)	Oats, garlic, onion, banana, chicory, dal	Fermented to SCFAs by beneficial bacteria
Probiotics (introduce bacteria)	Curd/dahi, lassi, kefir, kombucha, kimchi, miso	Temporary colonisation; modulate immune signalling

Diversity of plant foods	30 different plant foods/week (AM)	More plant species = more bacterial species
Fibre (25–38g/day)	Whole grains, legumes, vegetables, fruits	Primary substrate for gut bacteria
Polyphenols	Berries, green tea, turmeric, dark chocolate	Selectively feed Lactobacillus, Bifidobacterium

Module 2 (CS_B2) Key Takeaways

- Micronutrients are metabolic co-factors — without them, macronutrient metabolism is impaired
- Vitamin D3 deficiency affects 70–90% of Indians; supplement 2000–4000 IU daily with K2
- Non-haem iron absorption doubles when consumed with vitamin C; tea/coffee inhibit it
- Magnesium cofactors 300+ enzymes including all ATP reactions; 60–70% of people are deficient
- Antioxidant hormesis: mild exercise-induced ROS triggers adaptation; mega-dosing post-workout may blunt gains
- Gut microbiome diversity (30+ plant foods/week) improves absorption, recovery, and inflammation

1. Why do 70–90% of Indians have Vitamin D3 deficiency despite abundant sunshine?

2. Explain haem vs non-haem iron absorption and give three practical strategies to improve non-haem absorption.

3. What is the hormesis paradox in antioxidant nutrition for athletes?

4. Why is magnesium oxide a poor supplement choice compared to magnesium glycinate?

5. List three prebiotic foods and explain how they benefit the gut microbiome.